Concave Lens - HKDSE Physics Summary

1. Basic Properties of Concave Lens

1.1 Definition and Symbol

- **Concave Lens** = **Diverging Lens**
- Incident rays parallel to the principal axis **diverge** after refraction, appearing to come from the focus
- Symbol: Lens shape with inward curves on both ends

1.2 Concave Lens vs Convex Lens Comparison

Property	Convex Lens	Concave Lens
Type	Converging lens	Diverging lens
Light behavior	Parallel rays converge at focus	Parallel rays diverge, appearing from focus
Focal length sign	Positive (+ve)	Negative (-ve)

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2. Three Principal Rays for Concave Lens

Rule 1: Parallel in, Focus out

- A ray parallel to the principal axis **diverges** after refraction, with its extension passing through the focus F on the same side
- Mnemonic: **Parallel in, Focus out**

Rule 2: Focus in, Parallel out

- A ray directed towards the focus F' on the opposite side emerges parallel to the principal axis after refraction
- Mnemonic: **Focus in, Parallel out**

Rule 3: Straight through center

- A ray passing through the optical center C travels straight through without changing direction
- Mnemonic: **C in C out, Straight through**

3. Image Characteristics of Concave Lens

3.1 Nature of Image (Always Fixed)

Images formed by concave lenses **always** have the following characteristics:

- ✅ **Virtual** - Cannot be displayed on a screen
- ✅ **Erect** - Same orientation as the object
- ✅ **Diminished** - Smaller than the object
- ✅ **Position** - Between the lens and object, located between C and F

3.2 Important Notes

- Regardless of object position, the image characteristics of a concave lens **never change**
- All parallel incident rays after refraction by a concave lens must diverge from a focus on the focal plane

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4. Lens Formula and Magnification

4.1 Lens Formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Sign Convention:

- f
  = focal length (convex lens: +ve; **concave lens: -ve**)
- u
  = object distance (always +ve)
- v
  = image distance (real image: +ve; **virtual image: -ve**)

4.2 Linear Magnification

$M = \frac{v}{u} = \frac{h_i}{h_o}$

Where:

- M
  = magnification
- v
  = image distance
- u
  = object distance
- h_i
  = image height
- h_o
  = object height

Magnification characteristics of concave lens:

- Since
  v
  is negative (virtual image),
  M
  is also negative
- $|M| < 1$
  (diminished)

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5. Applications of Concave Lens

5.1 Peep-hole Lens

- Installed in doors for observation
- Uses concave lens to form diminished, erect virtual image
- Provides a wide field of view outside the door

5.2 Spectacles for Short-sightedness (Myopia)

- Problem with myopia: Light converges **in front of** the retina
- Concave lens diverges light, moving the focus back to the retina
- Corrects short-sightedness, allowing distant objects to be seen clearly

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6. Important Concept Summary

6.1 Absolute Rules of Concave Lens Imaging

1. **Image nature never changes** - Always virtual, erect, diminished
2. **Image position** - Always between object and lens
3. **Never forms real image** - Cannot be displayed on screen regardless of object distance

6.2 Common Misconceptions

❌ **Wrong**: When object is at focus, image is at infinity
✅ **Correct**: Image of concave lens is always between lens and object, never at infinity

❌ **Wrong**: Concave lens may form magnified or diminished images
✅ **Correct**: Image of concave lens is always diminished

❌ **Wrong**: Concave lens may form real or virtual images
✅ **Correct**: Image of concave lens is always virtual

6.3 Calculation Tips

1. **Focal length** f **must be negative** (concave lens)
2. **Image distance** v **result is negative** (virtual image)
3. **Magnification** M **take absolute value** to determine size

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7. HKDSE Exam Key Points

7.1 Must Remember

- Concave lens = Diverging lens
- Image nature: virtual, erect, diminished (all three essential)
- Focal length sign: negative

- Image distance sign: negative

7.2 Ray Diagram Points

1. Draw at least two rays to locate image position
2. Virtual image shown with dotted lines
3. Refraction occurs at the lens, distinguish incident and refracted rays clearly

7.3 Application Questions

- Myopia correction: Concave lens diverges light
- Peep-hole lens: Forms diminished, erect virtual image, expands field of view

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8. Quick Memory Mnemonics

Concave Lens Imaging Mnemonic:

> Concave lens, light diverges
> Virtual, erect, diminished - never changes
> Image between object and lens
> Short-sighted eyes, use it to correct

Three Rays Mnemonic:

> Parallel in, Focus out
> Focus in, Parallel out
> C in C out, Straight through

Study Suggestions:

1. Compare differences between convex and concave lenses
2. Memorize fixed image properties of concave lens
3. Practice using lens formula, pay attention to sign conventions
4. Understand applications of concave lens in daily life